

Taiwan Experience Education Program (TEEP): Research, Technology, Culture, and Life in Taiwan

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1. Introduction

This report presents my experience participating in the Taiwan Experience Education Program (TEEP), focusing on my research internship and my experiences living and exploring Taiwan. The main focus of my internship was the development and evaluation of an advanced Robot Nurse Helper system (RobotNurseHelper) using the Zenbo JrII social robot, integrating bilingual interaction, local LLM inference via Ollama (gemma3:1b), dynamic name substitution, and healthcare applications. The project included functions related to cataract surgery queries, object detection, and multi-state dialogue handling, as well as testing in a real-world environment, where we received valuable feedback for further improvement. In addition to research, I experienced Taiwan's academic and technological environment through activities such as the AIWAVE exhibition and a visit to Taipei 101. I also explored various cultural and natural destinations, including Longshan Temple, Chiang Kai-shek Memorial Hall, Kaohsiung, Shifen, Golden Waterfall, Tamsui, and Beitou. These experiences contributed to my technical knowledge, research skills, cultural understanding, and personal growth during the TEEP program.

2. Research Project: Robot Nurse Helper Using Zenbo JrII

The main focus of my internship was the development and optimization of a 24-state Robot Nurse Helper framework using the Zenbo JrII robotic platform. The project aimed to explore how artificial intelligence, local language models, and robotics could be applied in a healthcare environment to provide intelligent, context-aware assistance and improve interaction with patients. The project combined robotics, artificial intelligence, computer vision, local LLM inference (ollama with gemma3:1b), and human-robot interaction.

During the internship, I worked on several critical architectural components of the system:

- **Local LLM Integration & Optimization:** Implemented local model execution using Ollama with gemma3:1b for name extraction, intent recognition, and dynamic dialogue generation, ensuring responsive performance and avoiding external network bottlenecks.
- **Dynamic Name Substitution & Facial Expressions:** Developed robust runtime variable replacement (`{{PatientName}}`) and a content-aware facial expression selector (`GetDynamicFaceExpression`) that dynamically maps dialogue contexts to natural robot expressions (such as `HAPPY` for greetings, `INTERESTED` for medical/cataract diagnoses, `DOUBTING` for symptom check-ups, and `PLEASED` for correct quiz answers).

- **Bilingual Interaction & Healthcare Features:** Built bilingual English and Chinese interaction capabilities and integrated specialized modules for cataract surgery information queries and object detection using computer vision pipelines.

Working on this project gave me practical experience in integrating complex software backends, Qt multi-threading, C++ server logic, and protobuf communication protocols into a physical robotic platform. It also helped me understand that developing healthcare technology requires more than technical implementation; the system must be robust, reliable, and comfortable for the people who will interact with it.



3. Testing and Research Experience

One of the most valuable parts of my internship was evaluating and testing the Robot Nurse Helper system in a real-world environment. Before evaluation, much of the development, Qt AutoMoc compilation fixes, and state machine debugging were performed in a controlled laboratory setting. Real-world testing allowed us to observe how the robot performed when interacting under dynamic conditions.

The testing provided positive and encouraging feedback while also revealing areas where the system could be improved. This experience showed me the importance of user feedback and real-world evaluation. Some features that appear effective during laboratory testing may need adjustment when deployed in a live clinical or interactive environment.

The iterative process of development, testing, feedback, and system refinement became an important part of my research experience. Collaborating with professors and students from different backgrounds taught me how interdisciplinary teamwork combines healthcare knowledge with artificial intelligence and robotics to address real-world challenges.

4. Academic and Technology Experience in Taiwan

TEEP also gave me the opportunity to experience a different academic and technological environment. Working with professors and researchers in Taiwan helped me understand different approaches to research, project management, and collaborative software development.

One of the most interesting academic experiences was visiting the AIWAVE exhibition. As a computer science and AI student working on a robotics project, the exhibition was particularly meaningful. I was able to observe emerging technologies, computer vision applications, and different implementations of artificial intelligence. Seeing technologies beyond the university laboratory encouraged me to think about future applications of intelligent systems.



We also had the opportunity to visit Taipei 101 through an activity organized with our host institution. Besides being a memorable cultural experience, it provided an opportunity to connect with fellow researchers outside the laboratory environment and learn more about Taiwan's modern infrastructure and iconic architectural landmarks.

5. Life and Travel in Taiwan

Living in Taiwan was one of the most enjoyable parts of my TEEP experience. I had the opportunity to visit cultural landmarks such as Longshan Temple and Chiang Kai-shek Memorial Hall, where I learned more about Taiwanese traditions, historical milestones, and local culture. Outside the university, I visited many urban, natural, and historical destinations:

- **Taipei 101:** Explored Taiwan's renowned landmark, appreciating its architectural design and modern urban surroundings.



- **Kaohsiung:** Experienced southern Taiwan's vibrant port culture, modern urban development, and scenic waterfront attractions.



- **Shifen Waterfall & Golden Waterfall:** Showcased northern Taiwan's stunning natural landscapes and waterfalls.



- **Tamsui & Beitou:** Highlighted scenic riverside environments and rich local cultural heritage.



Traveling around Taiwan helped me become more independent and confident. Planning routes, navigating public transportation, managing time, and adapting to unfamiliar environments were valuable skills developed during these trips.

6. Challenges, Learning, and Personal Growth

Like any international experience, my internship involved challenges. Adapting to a new country, a different academic environment, and managing complex software integrations required flexibility, patience, and rigorous problem-solving. Debugging state machine logic, handling LLM history truncation to prevent token overflows, and ensuring real-time stability taught me resilience in software engineering.

These challenges became vital learning opportunities. Real-world testing taught me the importance of designing technology around actual users. Living abroad also encouraged me to step outside my comfort zone, communicate across cultures, and manage unexpected situations independently.

7. Conclusion

The Taiwan Experience Education Program was a valuable combination of research, technical development, practical experience, cultural exchange, and personal growth. Through the Robot Nurse Helper project on Zenbo JrII, I gained practical expertise in artificial intelligence, robotics, local LLM deployment, multi-threading, and system integration.

Beyond research, visiting the AIWAVE exhibition and exploring iconic destinations like Taipei 101 and Kaohsiung broadened my global perspective. I sincerely appreciate the guidance and support provided by my professors and Chang Gung University throughout the program, as well as the opportunity made possible through TEEP. The knowledge, technical skills, cultural understanding, and confidence I gained in Taiwan will deeply influence my future academic studies and professional career.